

WHITTIER, AK, USA

WMO: 702757

Lat: 60.777N

Lon: 148.719W

Elev: 12

StdP: 101.18

Time Zone: -9.00 (NAK)

Period: 94-05

WBAN: 26444

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-12.1	-21.0	0.6	-12.3	-18.0	0.8	-11.0	15.7	-7.6	15.2	-8.7	9.5	220	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	22.0	15.9	20.0	14.7	18.1	13.5	16.1	20.6	15.1	19.2	14.1	17.5	5.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.9	9.9	17.4	12.9	9.3	15.8	12.5	9.0	15.2	44.8	20.5	42.0	19.2	39.5	17.5	20.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 12.7	11.2	10.6		DB	-15.4	24.3	2.9	1.4	-17.5	25.2	-19.2	26.0	-20.8	26.8	-22.9	27.8
(5)				WB	-16.2	18.6	2.7	1.2	-18.2	19.4	-19.8	20.1	-21.3	20.8	-23.3	21.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-2.0	-1.1	-0.1	3.9	8.0	12.3	13.8	13.5	9.5	4.5	0.2	-1.3
	DBStd	6.72	4.80	4.40	3.61	2.71	2.69	2.79	2.17	2.35	2.24	2.85	3.87	4.20
	HDD10.0	2112	373	311	312	185	74	7	0	1	34	171	294	350
	HDD18.3	4822	631	544	571	434	320	181	144	151	264	429	544	608
	CDD10.0	335	0	0	0	1	12	77	117	109	19	0	0	0
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0
	CDH23.3	9	0	0	0	0	0	2	6	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.8	4.7	4.2	4.4	3.7	3.3	2.9	2.5	3.0	3.6	4.2	4.6	4.6
Precipitation	PrecAvg	4107	369	348	311	330	295	195	234	305	464	440	420	429
	PrecMax	4961	991	661	594	709	698	335	423	618	1011	916	1084	809
	PrecMin	2340	47	47	63	41	33	25	124	95	95	121	34	101
	PrecStd	584	210	170	156	161	195	76	82	146	216	180	282	179
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	5.0	7.4	14.9	17.6	22.8	23.9	22.6	17.2	11.3	7.9	6.3
		MCWB	3.1	3.5	5.1	9.7	11.1	16.4	16.8	16.6	11.9	10.1	6.6	5.0
	2%	DB	5.0	4.0	5.9	12.2	16.2	21.2	22.1	21.1	15.4	10.0	7.1	5.3
		MCWB	3.3	3.1	3.8	7.9	10.2	15.0	15.9	15.7	11.0	8.4	6.0	4.3
	5%	DB	3.9	3.0	4.5	9.5	14.1	19.1	19.4	19.4	14.0	8.9	5.9	4.0
		MCWB	2.8	2.5	2.8	6.2	9.3	13.6	14.5	14.6	10.2	7.6	4.6	3.2
	10%	DB	2.9	2.7	3.5	7.7	12.8	17.6	17.7	17.9	12.8	7.9	4.3	3.0
		MCWB	2.2	2.2	2.2	4.9	8.8	12.8	13.6	13.5	9.9	6.8	3.3	2.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	4.1	5.7	10.3	12.3	17.3	17.8	17.7	12.8	11.0	7.6	5.9
		MCDB	5.4	4.5	6.9	14.7	16.2	22.4	22.7	21.7	15.2	11.2	7.7	6.5
	2%	WB	3.7	3.4	4.3	8.1	11.1	15.5	16.0	15.9	12.0	9.0	6.1	4.7
		MCDB	4.4	3.7	5.2	11.2	14.9	20.1	20.6	20.0	13.6	9.2	6.9	5.4
	5%	WB	3.0	2.9	3.3	6.5	10.2	14.1	15.0	15.0	11.4	8.0	4.8	3.6
		MCDB	3.6	3.1	4.0	9.2	13.7	18.4	19.1	18.7	12.9	8.2	5.7	4.0
	10%	WB	2.5	2.5	2.7	5.3	9.2	13.1	14.1	14.1	10.9	7.5	3.8	2.8
		MCDB	3.0	2.7	3.3	7.2	12.3	17.1	17.1	17.3	12.3	7.8	4.5	3.2
Mean Daily Temperature Range	5% DB	MDBR	3.8	3.5	4.3	5.2	6.1	6.7	5.5	5.7	4.4	3.7	3.6	4.0
		MCDBR	4.0	3.2	5.0	7.9	8.6	9.8	8.8	8.7	6.1	4.3	4.1	4.0
		MCWBR	3.4	3.0	4.1	6.0	5.8	6.5	5.6	5.6	4.1	3.6	3.6	3.7
	5% WB	MCDBR	3.5	3.0	4.6	7.1	7.7	8.8	7.7	8.2	4.6	3.5	3.9	3.7
		MCWBR	3.2	2.9	4.0	5.7	5.8	6.2	5.4	5.5	3.7	3.4	3.5	3.6
Clear-Sky Solar Irradiance	taub	0.249	0.312	0.326	0.369	0.380	0.383	0.388	0.373	0.323	0.311	0.255	0.240	
	taud	2.054	2.063	2.074	2.017	2.115	2.243	2.306	2.346	2.474	2.266	2.040	1.914	
	Ebn at Noon	478	640	776	810	837	842	829	811	803	653	448	290	
	Edh at Noon	42	79	109	140	137	124	114	101	74	64	41	30	
All-Sky Solar Radiation	RadAvg	0.30	0.99	2.57	4.20	5.23	5.22	4.70	3.71	2.44	1.13	0.43	0.17	
	RadStd	0.04	0.15	0.30	0.43	0.45	0.28	0.37	0.41	0.35	0.14	0.07	0.02	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	N/A	+1.90	+1.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page